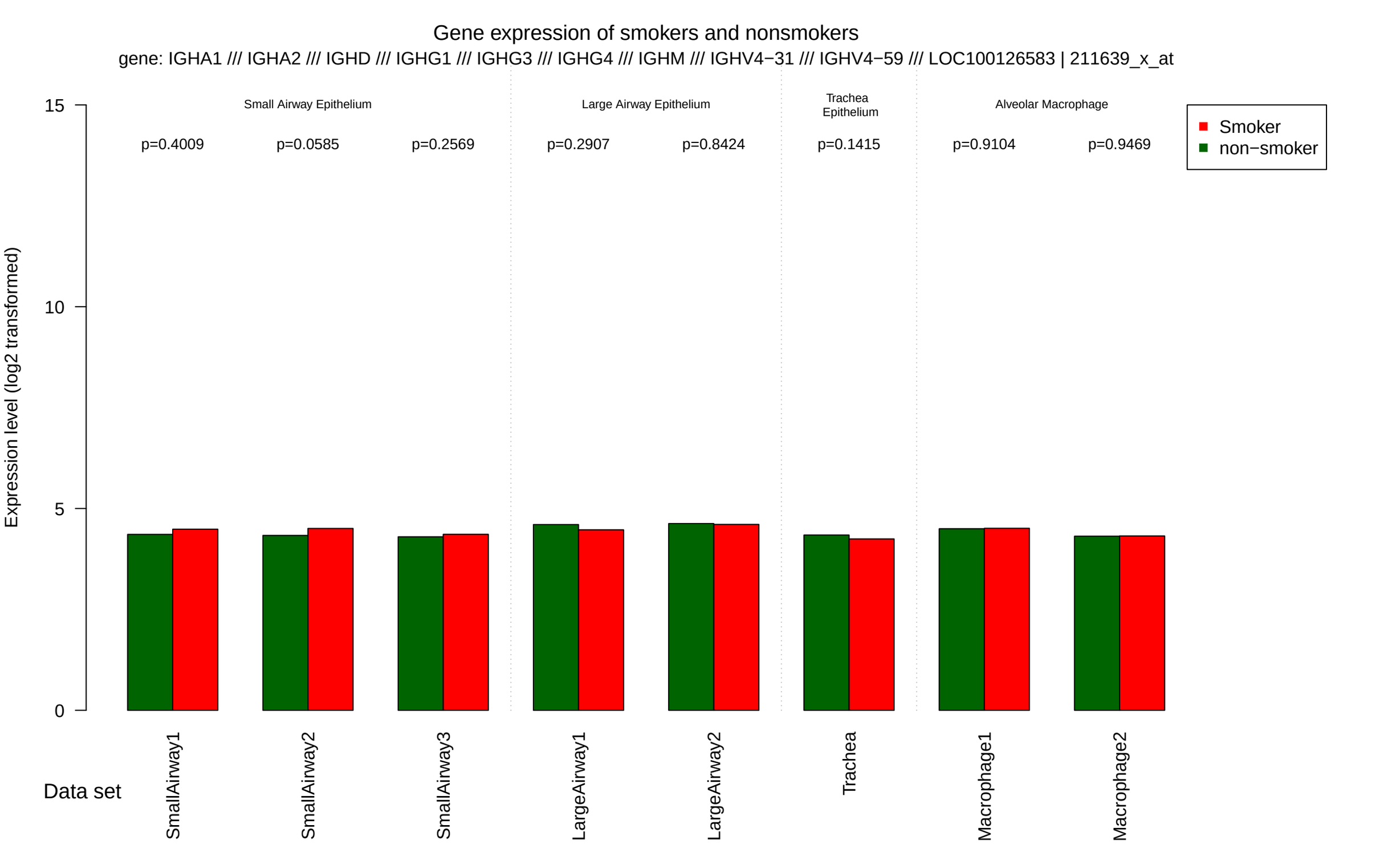
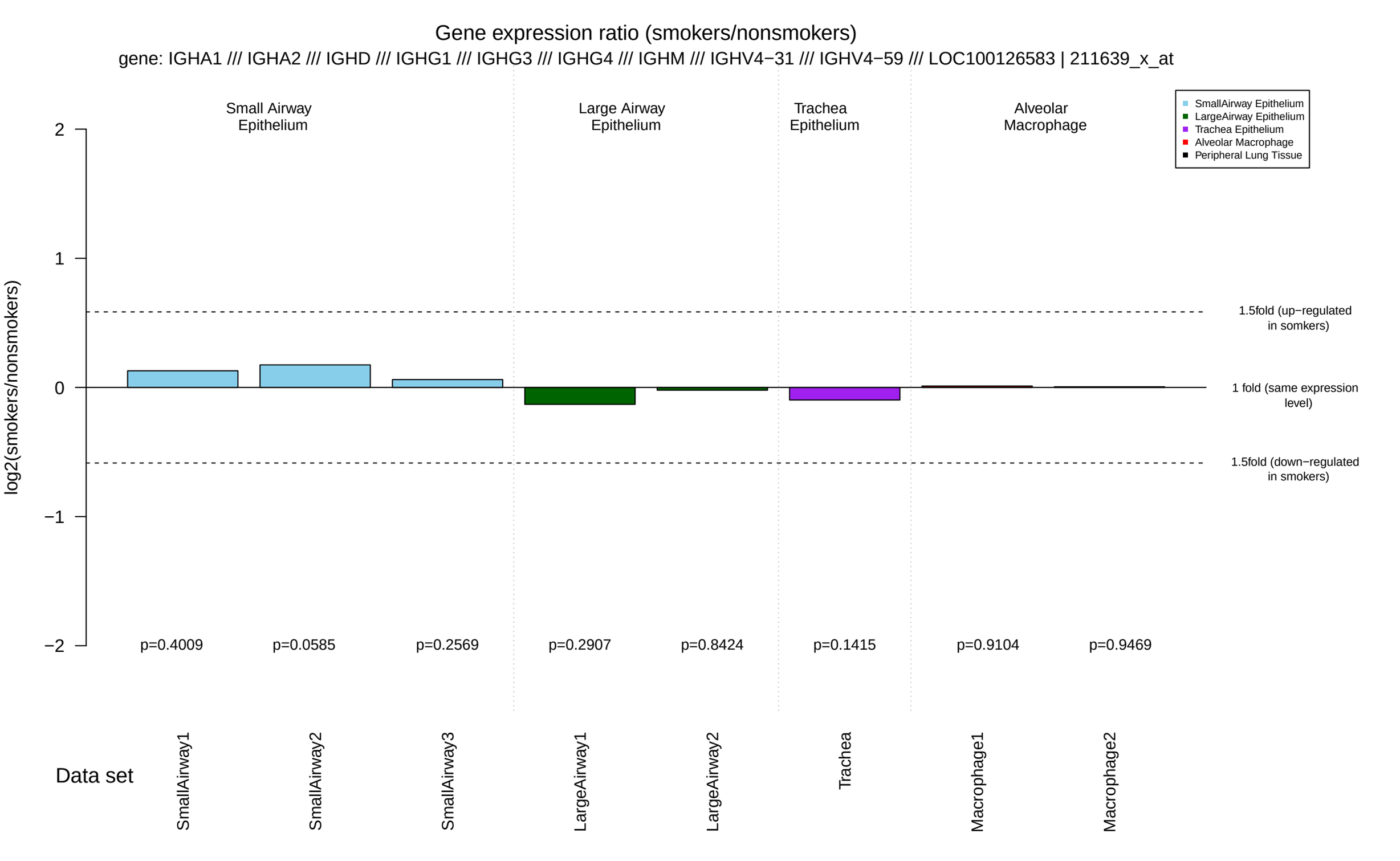
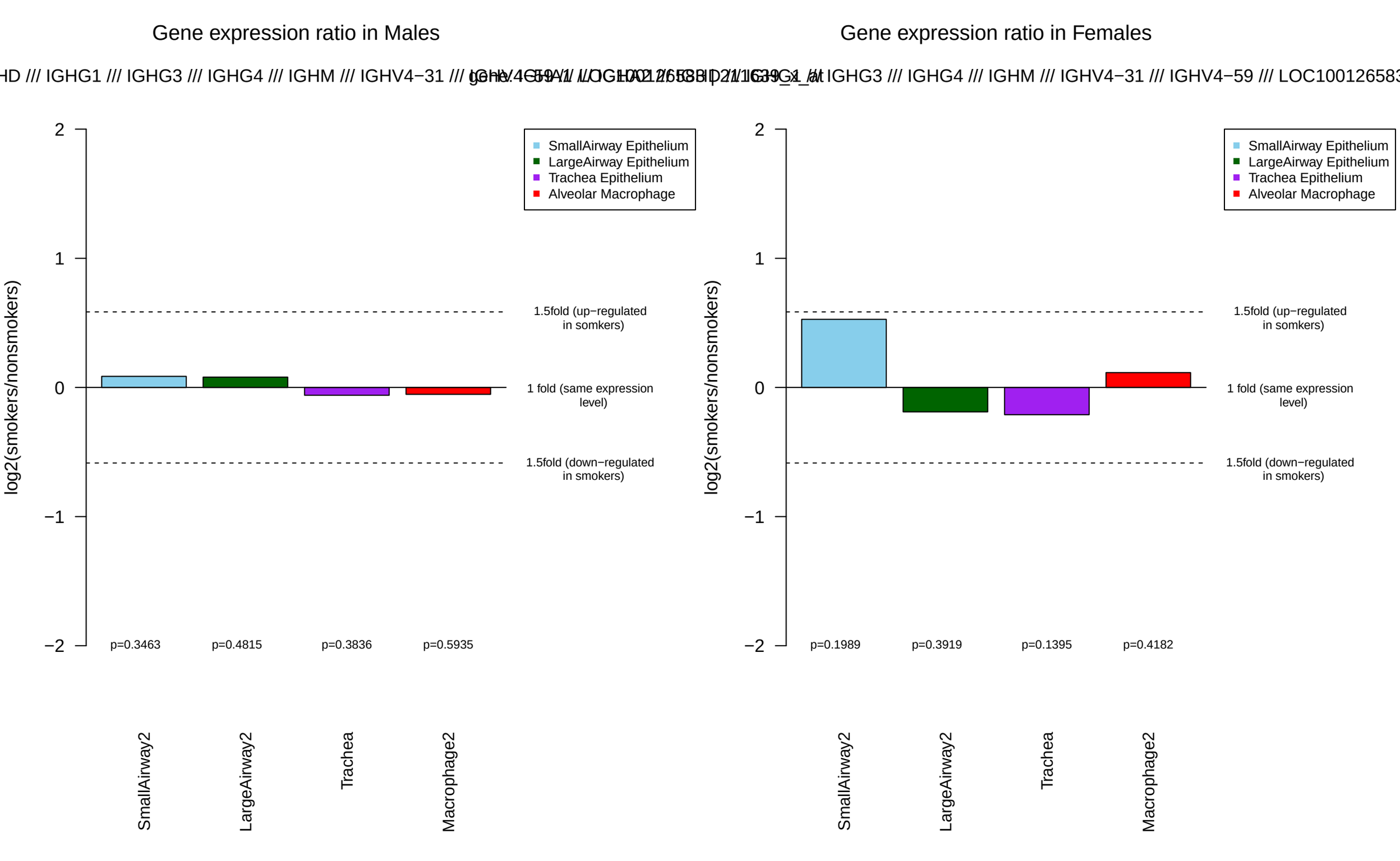
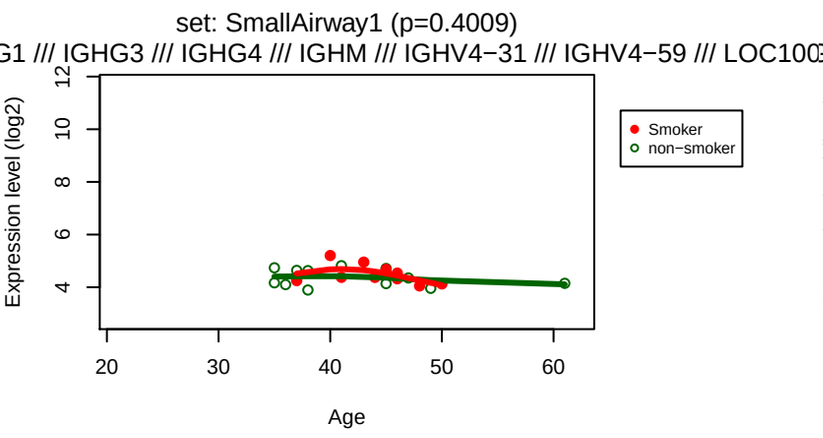
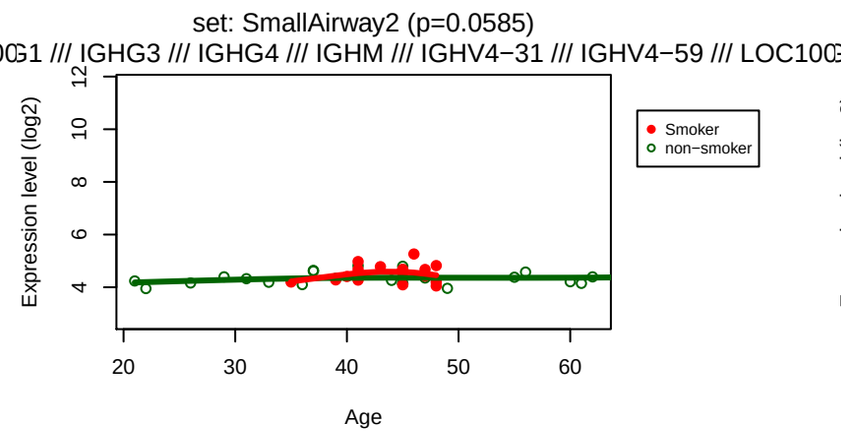
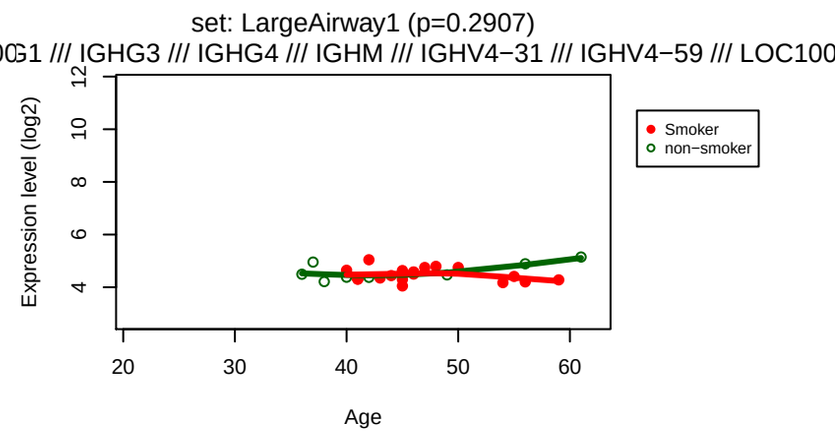

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583 | 211639_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4009
p=0.0585
p=0.2569
p=0.2907
p=0.8424
p=0.1415
p=0.9104
p=0.9469
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

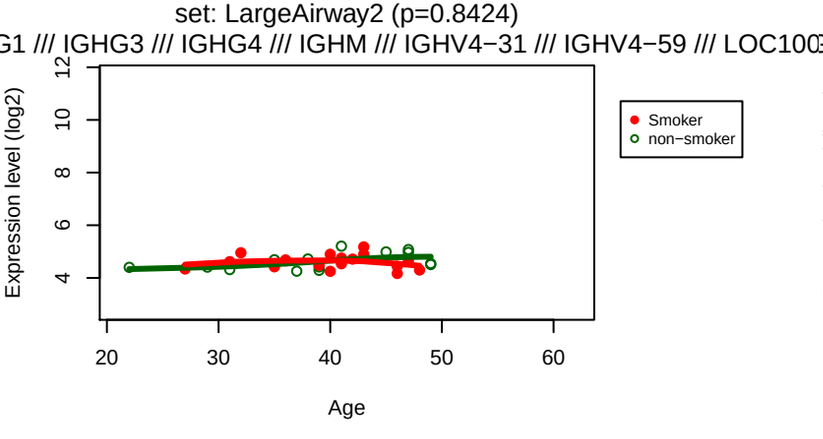

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583 | 211639_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4009
p=0.0585
p=0.2569
p=0.2907
p=0.8424
p=0.1415
p=0.9104
p=0.9469
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

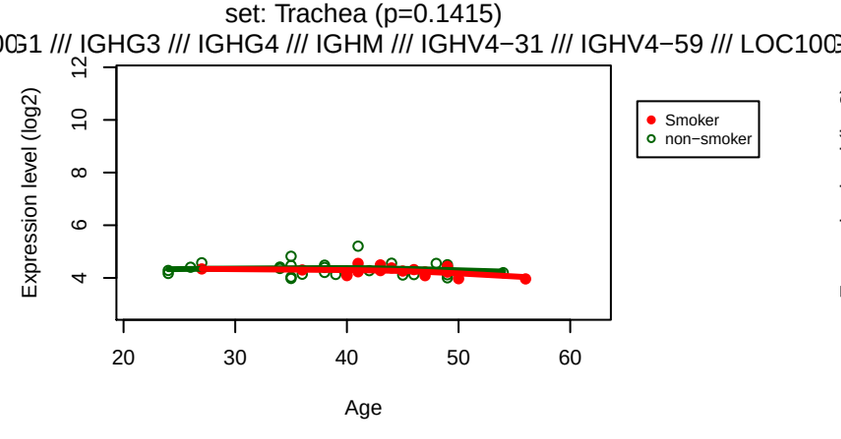

Gene expression ratio in Males
HD /// IGHG1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3463
p=0.4815
p=0.3836
p=0.5935
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
log2(smokers/nonsmokers)
p=0.1989
p=0.3919
p=0.1395
p=0.4182
SmallAirway2
LargeAirway2
Trachea
Macrophage2

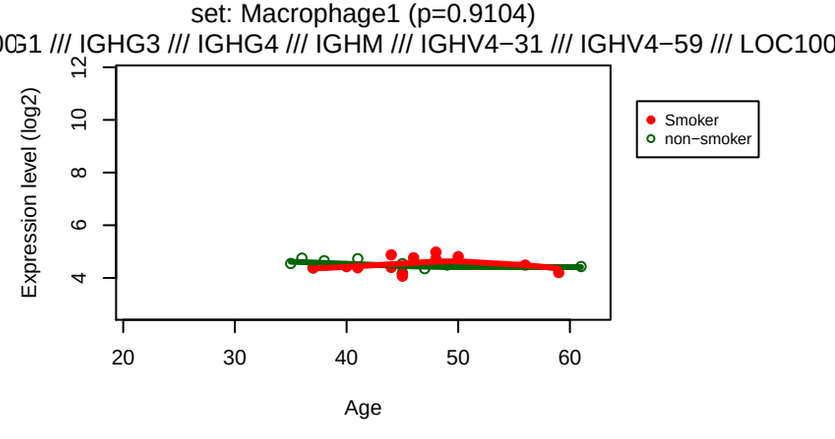

set: SmallAirway1 (p=0.4009)
G1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100
Expression level (log2)
Age
Smoker
non−smoker

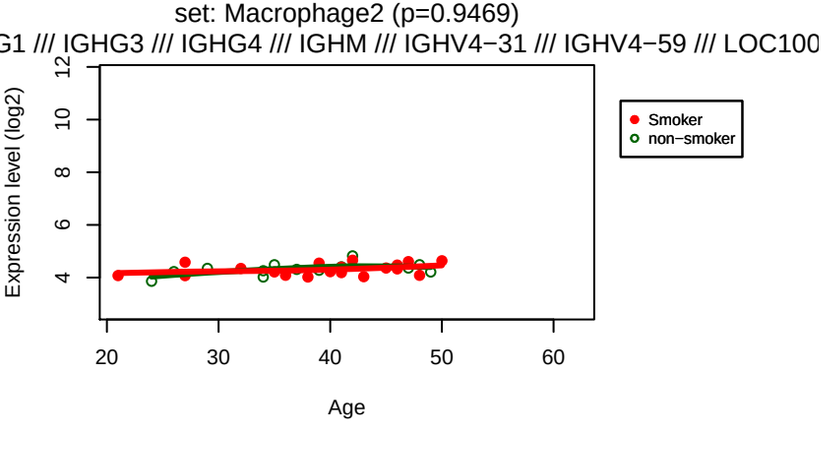

set: SmallAirway2 (p=0.0585)
31 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100
Expression level (log2)
Age
Smoker
non−smoker

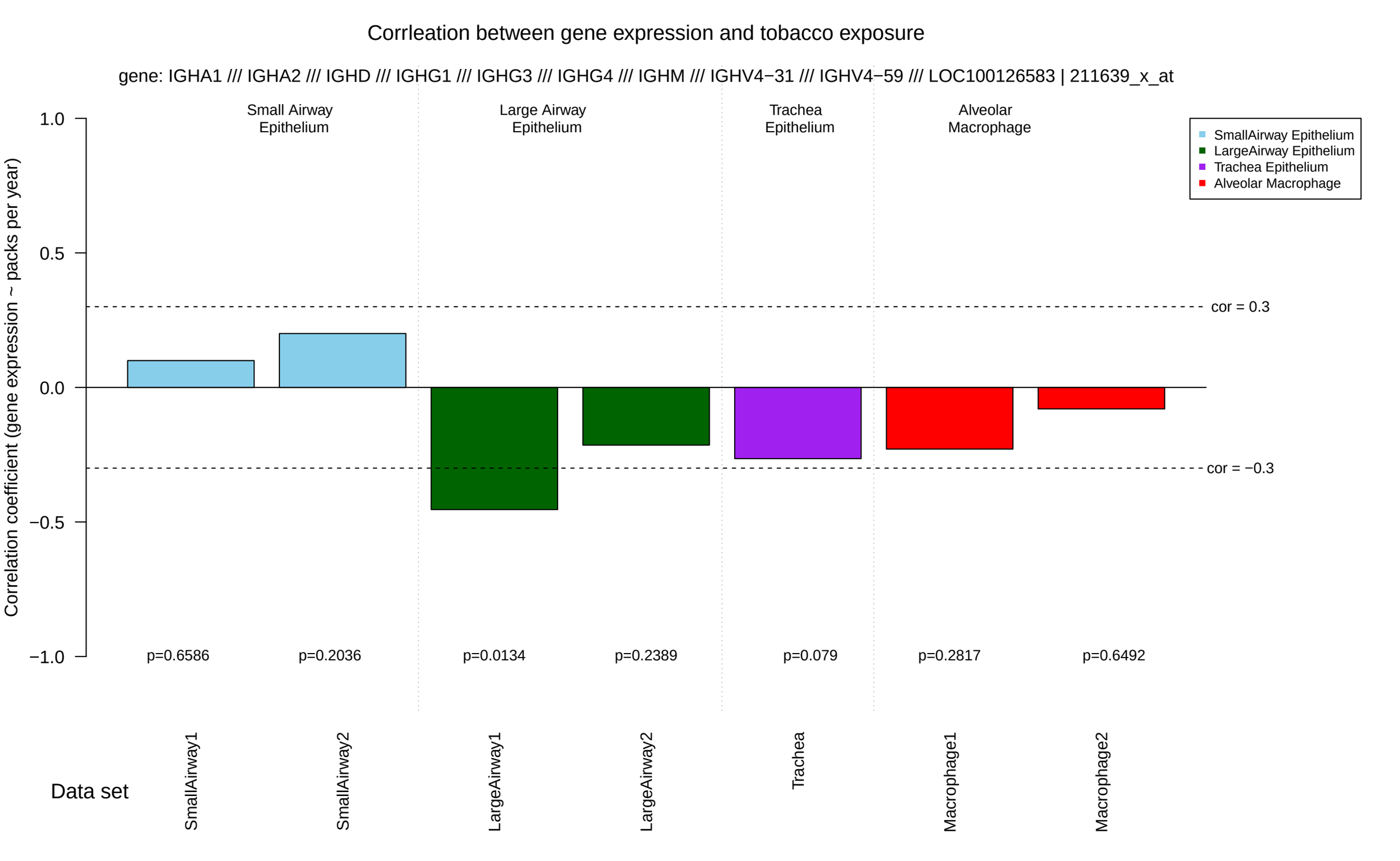

set: LargeAirway1 (p=0.2907)
31 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8424)
G1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1415)
31 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9104)
31 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9469)
G1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583 | 211639_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6586
p=0.2036
p=0.0134
p=0.2389
p=0.079
p=0.2817
p=0.6492
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2